

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1397	0.1903	$0.73\times$	0.1610	0.1781	$0.90\times$
DBQuery	15	13	0.1318	0.1269	$1.04\times$	0.1360	0.1728	$0.79\times$
DecisionTree	7	6	3.827	3.099	$1.24\times$	—	—	—
DomTree	15	14	0.1422	0.0629	$2.26\times$	0.0585	0.0814	$0.72\times$
KDTree	17	16	0.1764	0.1449	$1.22\times$	—	—	—
LinearListReduction	11	11	0.0308	0.0064	$4.81\times$	—	—	—
List	2	2	0.1302	0.1113	$1.17\times$	0.3754	0.3391	$1.11\times$
MonoTree	3	2	0.0179	0.0226	$0.79\times$	0.0924	0.0803	$1.15\times$
ObjectGraph	5	4	0.0898	0.0945	$0.95\times$	0.1482	0.1607	$0.92\times$
OctTree_sumMass	16	—	0.0362	0.0295	$1.23\times$	—	—	—
OctTree_sumEnergy	16	—	0.0373	0.0367	$1.02\times$	—	—	—
OctTree_countActive	16	—	0.0369	0.0338	$1.09\times$	—	—	—
OctTree_countParticles	16	—	0.0370	0.0357	$1.04\times$	—	—	—
OctTree_barnesHutPotential	16	—	0.0412	0.0412	$1.00\times$	—	—	—
OctTree_fmmPotential	16	—	0.0967	0.1193	$0.81\times$	—	—	—
OctTree_scaleEnergy	16	—	—	—	—	0.1530	0.1782	$0.86\times$
OctTree_clearFlags	16	—	—	—	—	0.1554	0.1780	$0.87\times$
PiecewiseFunctions	8	7	0.1395	0.0966	$1.44\times$	0.2406	0.2602	$0.92\times$
TernaryTree	5	3	0.0277	0.0270	$1.02\times$	0.0847	0.0898	$0.94\times$
Trie	10	9	0.0630	0.0682	$0.92\times$	0.1522	0.1761	$0.86\times$
ColorOctree	25	18	0.0946	0.0813	$1.16\times$	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. $>1\times$ means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	$1.65\times$	$0.52\times$	$2.52\times$	$0.89\times$	$8.58\times$	$0.41\times$	$8.28\times$
DBQuery	$1.03\times$	$0.47\times$	$1.22\times$	$0.36\times$	$7.90\times$	$0.22\times$	$5.77\times$
DecisionTree	$1.16\times$	$0.59\times$	$16.61\times$	$3.00\times$	$0.86\times$	$3.21\times$	$36.11\times$
DomTree	$1.54\times$	$0.57\times$	$1.81\times$	$0.85\times$	$16.13\times$	$0.23\times$	$10.29\times$
KDTree	$1.27\times$	$0.77\times$	$1.52\times$	$0.98\times$	$0.23\times$	$0.95\times$	$4.84\times$
LinearListReduction	$5.55\times$	$0.90\times$	$1.26\times$	$2.02\times$	$4.19\times$	$2.94\times$	$10.84\times$
List	$0.81\times$	$0.80\times$	$2.52\times$	$0.49\times$	$0.22\times$	$1.00\times$	$1.45\times$
MonoTree	$0.98\times$	$0.85\times$	$0.33\times$	$0.83\times$	$0.21\times$	$0.85\times$	$1.24\times$
ObjectGraph	$0.95\times$	$0.75\times$	$5.10\times$	$0.82\times$	$0.48\times$	$0.78\times$	$2.85\times$
OctTree_sumMass	$1.30\times$	$0.89\times$	$3.75\times$	$1.14\times$	$0.31\times$	$1.22\times$	$4.87\times$
OctTree_sumEnergy	$1.03\times$	$0.87\times$	$0.09\times$	$1.19\times$	$0.03\times$	$1.50\times$	$2.52\times$
OctTree_countActive	$1.14\times$	$1.01\times$	$0.97\times$	$1.43\times$	$0.19\times$	$1.89\times$	$20.71\times$
OctTree_countParticles	$1.24\times$	$1.00\times$	$2.71\times$	$1.50\times$	$0.51\times$	$2.23\times$	$53.92\times$
OctTree_barnesHutPotential	$1.03\times$	$0.82\times$	$0.04\times$	$0.87\times$	$0.04\times$	$0.79\times$	$2.84\times$
OctTree_fmmPotential	$0.98\times$	$0.87\times$	$0.00\times$	$0.45\times$	$0.02\times$	$0.36\times$	$2.13\times$
OctTree_scaleEnergy	$0.81\times$	$0.49\times$	$0.44\times$	$0.37\times$	$0.22\times$	$0.46\times$	$2.24\times$
OctTree_clearFlags	$0.82\times$	$0.46\times$	$0.44\times$	$0.53\times$	$0.22\times$	$0.58\times$	$2.31\times$
PiecewiseFunctions	$1.14\times$	$0.60\times$	$2.98\times$	$0.88\times$	$2.31\times$	$0.44\times$	$4.44\times$
TernaryTree	$1.13\times$	$0.79\times$	$0.38\times$	$0.53\times$	$0.21\times$	$0.75\times$	$1.40\times$
Trie	$1.12\times$	$0.56\times$	$2.96\times$	$0.66\times$	$3.15\times$	$0.18\times$	$6.46\times$
ColorOctree	$1.18\times$	$0.84\times$	$2.00\times$	$0.83\times$	$0.25\times$	$0.85\times$	$4.41\times$
Geomean	$1.18\times$	$0.71\times$	$0.92\times$	$0.85\times$	$0.48\times$	$0.77\times$	$4.93\times$

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.5e+08/ 5.7e+08	9.9e+08 /1.9e+09	1.3e+07/ 5.2e+06	2.8e+05 /3.2e+05	1.3e+06/ 1.5e+05	1.1e+04 /2.7e+04	5.1e+07/ 6.2e+06
DBQuery	9.2e+08/ 8.9e+08	1.1e+09 /2.2e+09	3.9e+06/ 3.2e+06	1.6e+05 /4.6e+05	8.1e+05/ 1.0e+05	7.8e+03 /3.5e+04	2.0e+07/ 3.5e+06
DecisionTree	1.8e+10/ 1.6e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.3e+07	1.3e+05/ 4.2e+04	1.8e+06 /2.1e+06	1.3e+05/ 3.9e+04	1.1e+09/ 3.0e+07
DomTree	8.4e+08/ 5.5e+08	1.2e+09 /2.2e+09	1.2e+07/ 6.5e+06	1.0e+05 /1.2e+05	1.1e+06/ 6.6e+04	7.2e+03 /3.2e+04	5.7e+07/ 5.6e+06
KDTree	8.9e+08/ 7.0e+08	3.1e+09 /4.0e+09	1.4e+07/ 9.1e+05	1.7e+03 /1.8e+03	2.4e+04 /1.0e+05	1.5e+03 /1.6e+03	4.5e+07/ 9.2e+06
LinearListReduction	1.4e+08/ 2.6e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	6.4e+02/ 3.2e+02	2.8e+05/ 6.7e+04	5.2e+02/ 1.8e+02	1.2e+07/ 1.1e+06
List	1.2e+09 /1.5e+09	4.5e+09 /5.6e+09	9.9e+06/ 3.9e+06	2.8e+05 /5.7e+05	4.7e+04 /2.1e+05	1.4e+04 /1.4e+04	4.8e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.6e+05 /7.9e+05	4.0e+04 /4.9e+04	4.1e+03 /2.0e+04	2.8e+03 /3.3e+03	2.2e+06/ 1.8e+06
ObjectGraph	8.3e+08 /8.8e+08	3.0e+09 /4.0e+09	9.9e+06/ 1.9e+06	2.0e+05 /2.4e+05	3.1e+04 /6.4e+04	1.2e+04 /1.6e+04	2.5e+07/ 8.6e+06
OctTree_sumMass	1.8e+08/ 1.4e+08	5.7e+08 /6.3e+08	7.3e+05/ 1.9e+05	5.1e+02/ 4.5e+02	2.6e+03 /8.5e+03	3.9e+02/ 3.2e+02	6.1e+06/ 1.3e+06
OctTree_sumEnergy	1.9e+08/ 1.8e+08	7.7e+08 /8.9e+08	3.4e+05 /3.7e+06	6.0e+02/ 5.0e+02	2.1e+03 /8.3e+04	5.0e+02/ 3.3e+02	6.2e+06/ 2.5e+06
OctTree_countActive	4.1e+08/ 3.6e+08	5.8e+08 / 5.8e+08	8.9e+05 /1.4e+06	4.6e+02 / 4.6e+02	4.3e+03 /2.3e+04	5.2e+02 / 2.7e+02	6.2e+06 / 3.1e+05
OctTree_countParticles	1.9e+08/ 1.5e+08	5.7e+08/ 5.7e+08	8.2e+05/ 3.0e+05	6.2e+02/ 4.2e+02	4.1e+03 /8.0e+03	4.8e+02/ 2.1e+02	6.1e+06/ 1.1e+05
OctTree_barnesHutPotential	2.1e+08/ 2.0e+08	8.2e+08 /1.0e+09	6.4e+04 /1.5e+06	4.6e+02 /5.3e+02	1.6e+03 /4.1e+04	2.9e+02 /3.7e+02	6.2e+06/ 2.2e+06
OctTree_fmmPotential	4.9e+08 /5.0e+08	1.6e+09 /1.8e+09	2.5e+04 /4.9e+06	5.2e+02 /1.1e+03	1.6e+03 /9.8e+04	3.5e+02 /9.8e+02	4.5e+06/ 2.1e+06
OctTree_scaleEnergy	3.6e+08 /4.4e+08	9.1e+08 /1.9e+09	1.1e+06 /2.6e+06	1.4e+05 /3.7e+05	8.8e+03 /4.0e+04	5.2e+03 /1.1e+04	5.6e+06/ 2.5e+06
OctTree_clearFlags	3.6e+08 /4.5e+08	8.0e+08 /1.7e+09	1.2e+06 /2.7e+06	2.2e+05 /4.1e+05	9.8e+03 /4.5e+04	6.2e+03 /1.1e+04	5.6e+06/ 2.4e+06
PiecewiseFunctions	1.3e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.4e+07/ 4.7e+06	4.0e+05 /4.6e+05	2.1e+05/ 9.0e+04	2.0e+04 /4.4e+04	4.5e+07/ 1.0e+07
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	5.3e+05 /1.4e+06	6.6e+04 /1.2e+05	4.6e+03 /2.2e+04	2.4e+03 /3.1e+03	3.4e+06/ 2.4e+06
Trie	6.5e+08/ 5.8e+08	1.2e+09 /2.2e+09	8.3e+06/ 2.8e+06	1.9e+05 /2.8e+05	1.6e+05/ 5.2e+04	8.0e+03 /4.4e+04	2.6e+07/ 4.0e+06
ColorOctree	4.8e+08/ 4.0e+08	1.6e+09 /2.0e+09	9.5e+06/ 4.7e+06	7.5e+02 /9.0e+02	2.7e+04 /1.1e+05	6.2e+02 /7.3e+02	1.8e+07/ 4.2e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.3006	1.116	0.3685	<i>OOM</i>
DBQuery	0.2679	0.3365	0.2997	1.115
DecisionTree	3.827	5.671	3.099	25.35
DomTree	0.2007	0.3539	0.1444	0.5269
KDTree	0.1764	0.2442	0.1449	0.8727
LinearListReduction	0.0308	0.1337	0.0064	<i>OOM</i>
List	0.5056	<i>OOM</i>	0.4504	<i>OOM</i>
MonoTree	0.1103	0.1829	0.1029	0.3982
ObjectGraph	0.2380	0.5080	0.2552	0.7277
OctTree_sumMass	0.0362	0.0500	0.0295	<i>OOM</i>
OctTree_sumEnergy	0.0373	0.0583	0.0367	<i>OOM</i>
OctTree_countActive	0.0369	0.0631	0.0338	<i>OOM</i>
OctTree_countParticles	0.0370	0.0573	0.0357	<i>OOM</i>
OctTree_barnesHutPotential	0.0412	0.0633	0.0412	<i>OOM</i>
OctTree_fmmPotential	0.0967	0.0984	0.1193	<i>OOM</i>
OctTree_scaleEnergy	0.1530	0.1539	0.1782	<i>OOM</i>
OctTree_clearFlags	0.1554	0.1527	0.1780	<i>OOM</i>
PiecewiseFunctions	0.3801	0.6938	0.3568	2.166
TernaryTree	0.1124	0.1624	0.1169	0.6181
Trie	0.2152	0.3253	0.2443	1.042
ColorOctree	0.0946	0.1220	0.0813	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	0.82 $\times$	3.71$\times$	3.03$\times$	–
DBQuery	0.89 $\times$	1.26$\times$	1.12$\times$	0.30 $\times$
DecisionTree	1.24$\times$	1.48$\times$	1.83$\times$	0.22 $\times$
DomTree	1.39$\times$	1.76$\times$	2.45$\times$	0.67 $\times$
KDTree	1.22$\times$	1.38$\times$	1.69$\times$	0.28 $\times$
LinearListReduction	4.81$\times$	4.34$\times$	20.91$\times$	–
List	1.12$\times$	–	–	–
MonoTree	1.07 $\times$	1.66$\times$	1.78$\times$	0.46 $\times$
ObjectGraph	0.93 $\times$	2.13$\times$	1.99$\times$	0.70 $\times$
OctTree_sumMass	1.23$\times$	1.38$\times$	1.69$\times$	–
OctTree_sumEnergy	1.02 $\times$	1.57$\times$	1.59$\times$	–
OctTree_countActive	1.09 $\times$	1.71$\times$	1.87$\times$	–
OctTree_countParticles	1.04 $\times$	1.55$\times$	1.60$\times$	–
OctTree_barnesHutPotential	1.00 $\times$	1.53$\times$	1.54$\times$	–
OctTree_fmmPotential	0.81 $\times$	1.02 $\times$	0.82 $\times$	–
OctTree_scaleEnergy	0.86 $\times$	1.01 $\times$	0.86 $\times$	–
OctTree_clearFlags	0.87 $\times$	0.98 $\times$	0.86 $\times$	–
PiecewiseFunctions	1.07 $\times$	1.83$\times$	1.94$\times$	0.32 $\times$
TernaryTree	0.96 $\times$	1.44$\times$	1.39$\times$	0.26 $\times$
Trie	0.88 $\times$	1.51$\times$	1.33$\times$	0.31 $\times$
ColorOctree	1.16$\times$	1.29$\times$	1.50$\times$	–

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0122	0.0799	0.0124	<i>OOM</i>	0.98 $\times$	6.54$\times$	6.43$\times$	–
branchStatsPass	F	2/9	78%	0.0119	0.0804	0.0182	<i>OOM</i>	0.65 $\times$	6.76$\times$	4.42$\times$	–
castInstCountPass	F	3/9	67%	0.0183	0.0803	0.0084	<i>OOM</i>	2.17$\times$	4.39$\times$	9.51$\times$	–
has cycle	F	4/9	56%	0.0348	0.0934	0.0939	<i>OOM</i>	0.37 $\times$	2.69$\times$	0.99 $\times$	–
instCountPass	F	2/9	78%	0.0124	0.0797	0.0120	<i>OOM</i>	1.04 $\times$	6.43$\times$	6.66$\times$	–
latencyModelPass	F	3/9	67%	0.0121	0.0845	0.0126	<i>OOM</i>	0.96 $\times$	6.97$\times$	6.68$\times$	–
memoryOpStatsPass	F	3/9	67%	0.0120	0.0803	0.0213	<i>OOM</i>	0.56 $\times$	6.70$\times$	3.77$\times$	–
stripSideEffectsPass	M	7/9	22%	0.0803	0.1838	0.0907	<i>OOM</i>	0.89 $\times$	2.29$\times$	2.03$\times$	–
targetReturnPass	M	9/9	0%	0.0807	0.2010	0.0874	<i>OOM</i>	0.92 $\times$	2.49$\times$	2.30$\times$	–
throughputModelPass	F	3/9	67%	0.0122	0.0806	0.0030	<i>OOM</i>	4.12$\times$	6.59$\times$	27.15$\times$	–
verifyIR	F	9/9	0%	0.0137	0.0719	0.0085	<i>OOM</i>	1.62$\times$	5.23$\times$	8.48$\times$	–
Total				0.3006	1.116	0.3685	–	0.82 $\times$	3.71$\times$	3.03$\times$	–
Geomean								1.04 $\times$	4.79$\times$	5.00$\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.2e+07/ 9.8e+06	5.1e+07/ 4.5e+07	1.2e+06/ 1.0e+05	2.9e+02/ 9.9e+01	2.2e+05/ 2.7e+03	2.6e+02/ 7.1e+01	4.8e+06/ 1.3e+04
branchStatsPass	F	6.0e+07/ 1.4e+07	7.9e+07 /8.6e+07	8.1e+05/ 2.8e+05	1.8e+02/ 1.5e+02	5.2e+04/ 1.2e+04	1.5e+02/ 1.1e+02	4.7e+06/ 3.6e+05
castInstCountPass	F	9.3e+07/ 6.7e+06	5.1e+07/ 3.9e+07	1.4e+04 /1.0e+05	2.0e+02/ 8.3e+01	9.2e+02 /2.7e+03	1.8e+02/ 6.9e+01	4.7e+06/ 9.2e+03
has cycle	F	1.3e+08/ 8.2e+07	1.1e+08 /1.5e+08	2.4e+06/ 6.1e+05	8.5e+04/ 5.7e+04	1.2e+05/ 2.3e+04	2.1e+03/ 6.4e+02	5.0e+06/ 1.2e+06
instCountPass	F	6.1e+07/ 1.2e+07	5.6e+07 /5.7e+07	1.2e+06/ 1.0e+05	3.1e+02/ 1.1e+02	2.3e+05/ 2.7e+03	2.8e+02/ 9.0e+01	4.7e+06/ 3.3e+04
latencyModelPass	F	6.1e+07/ 1.0e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.3e+04	2.3e+02/ 1.0e+02	2.0e+05/ 4.1e+03	2.2e+02/ 8.7e+01	4.8e+06/ 4.2e+05
memoryOpStatsPass	F	6.1e+07/ 1.7e+07	8.5e+07 /9.1e+07	6.6e+05 /8.0e+05	1.8e+02/ 1.3e+02	3.2e+04/ 2.8e+04	1.7e+02/ 1.2e+02	4.7e+06/ 3.2e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.9e+06/ 1.2e+06	1.0e+05 /1.3e+05	6.5e+03 /2.1e+04	3.1e+03 /6.4e+03	4.3e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /1.9e+08	1.7e+08 /6.3e+08	1.4e+06/ 1.2e+06	9.0e+04 /1.2e+05	5.1e+03 /2.3e+04	3.5e+03 /1.8e+04	3.8e+06/ 1.5e+06
throughputModelPass	F	6.2e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.4e+04	3.9e+02/ 1.0e+02	2.0e+05/ 3.6e+03	3.2e+02/ 8.2e+01	4.8e+06/ 6.9e+05
verifyIR	F	5.5e+07/ 1.6e+07	7.4e+07 /8.6e+07	1.1e+06/ 7.9e+05	1.1e+03 /1.4e+03	2.1e+05/ 2.8e+04	1.0e+03 /1.3e+03	4.7e+06/ 4.1e+05
Total		9.5e+08/5.7e+08	9.9e+08/1.9e+09	1.3e+07/5.2e+06	2.8e+05/3.2e+05	1.3e+06/1.5e+05	1.1e+04/2.7e+04	5.1e+07/6.2e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/−	1.6e+08/−	2.3e+06/−	5.4e+02/−	9.3e+04/−	5.1e+02/−	1.4e+07/−
branchStatsPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	5.3e+02/−	4.4e+04/−	5.2e+02/−	1.4e+07/−
castInstCountPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	4.3e+02/−	4.6e+04/−	4.0e+02/−	1.4e+07/−
has cycle	F	4.7e+08/−	2.1e+08/−	2.5e+06/−	4.6e+02/−	7.1e+04/−	4.6e+02/−	1.7e+07/−
instCountPass	F	4.0e+08/−	1.6e+08/−	2.3e+06/−	4.6e+02/−	9.9e+04/−	4.5e+02/−	1.4e+07/−
latencyModelPass	F	4.3e+08/−	1.7e+08/−	2.1e+06/−	5.8e+02/−	4.7e+04/−	5.6e+02/−	1.4e+07/−
memoryOpStatsPass	F	4.1e+08/−	1.9e+08/−	2.1e+06/−	4.1e+02/−	4.3e+04/−	3.9e+02/−	1.4e+07/−
stripSideEffectsPass	M	6.6e+08/−	3.3e+08/−	8.1e+05/−	1.0e+05/−	1.6e+04/−	1.0e+04/−	2.4e+07/−
targetReturnPass	M	6.5e+08/−	3.2e+08/−	1.1e+06/−	1.2e+05/−	1.9e+04/−	2.1e+04/−	2.2e+07/−
throughputModelPass	F	4.1e+08/−	1.7e+08/−	2.1e+06/−	4.6e+02/−	4.7e+04/−	4.2e+02/−	1.4e+07/−
verifyIR	F	3.6e+08/−	1.2e+08/−	2.5e+06/−	6.7e+02/−	2.1e+05/−	5.8e+02/−	1.1e+07/−
Total		5.0e+09/−	2.2e+09/−	2.2e+07/−	2.2e+05/−	7.4e+05/−	3.5e+04/−	1.7e+08/−

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0678	0.0755	0.0861	0.2971	0.79 $\times$	1.11$\times$	0.88 $\times$	0.25 $\times$
countJoins	F	3/15	80%	0.0186	0.0282	0.0117	0.0852	1.59$\times$	1.52$\times$	2.42$\times$	0.33 $\times$
filterSelectivitySkew	F	5/15	67%	0.0186	0.0279	0.0125	0.0858	1.49$\times$	1.50$\times$	2.24$\times$	0.33 $\times$
hashJoinPressure	F	5/15	67%	0.0189	0.0321	0.0127	0.0861	1.50$\times$	1.69$\times$	2.53$\times$	0.37 $\times$
scaleCosts	M	15/15	0%	0.0682	0.0756	0.0867	0.2969	0.79 $\times$	1.11$\times$	0.87 $\times$	0.25 $\times$
sumCost	F	6/15	60%	0.0296	0.0395	0.0428	0.0897	0.69 $\times$	1.34$\times$	0.92 $\times$	0.44 $\times$
sumMemory	F	6/15	60%	0.0185	0.0283	0.0152	0.0879	1.21$\times$	1.53$\times$	1.86$\times$	0.32 $\times$
sumRows	F	6/15	60%	0.0276	0.0294	0.0321	0.0866	0.86 $\times$	1.07 $\times$	0.92 $\times$	0.34 $\times$
Total				0.2679	0.3365	0.2997	1.115	0.89 $\times$	1.26$\times$	1.12$\times$	0.30 $\times$
Geomean								1.06 $\times$	1.34$\times$	1.42$\times$	0.33 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.6e+08 /2.2e+08	1.9e+08 /7.2e+08	2.5e+05 /1.4e+06	8.7e+04 /2.3e+05	3.2e+04/ 2.9e+04	2.5e+03 /1.8e+04	2.3e+06/ 9.4e+05
countJoins	F	9.4e+07/ 5.9e+07	7.4e+07/ 7.1e+07	6.0e+05/ 2.5e+04	3.9e+02/ 2.1e+02	1.4e+05/ 2.0e+03	3.5e+02/ 1.6e+02	2.7e+06/ 5.8e+04
filterSelectivitySkew	F	9.4e+07/ 6.3e+07	1.0e+08 /1.1e+08	5.1e+05/ 8.0e+04	2.2e+02/ 2.2e+02	1.2e+05/ 6.5e+03	2.1e+02/ 2.0e+02	2.7e+06/ 3.3e+05
hashJoinPressure	F	9.6e+07/ 6.4e+07	8.4e+07 /9.9e+07	7.3e+05/ 1.0e+05	2.5e+02/ 1.4e+02	1.9e+05/ 7.5e+03	2.2e+02/ 1.2e+02	2.7e+06/ 1.4e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.6e+05 /1.3e+06	7.5e+04 /2.3e+05	3.6e+04/ 2.8e+04	3.6e+03 /1.6e+04	2.5e+06/ 1.1e+06
sumCost	F	8.2e+07/ 7.5e+07	9.0e+07 /1.2e+08	5.8e+05/ 7.7e+04	5.8e+02/ 3.5e+02	1.2e+05/ 1.1e+04	4.2e+02/ 2.6e+02	2.4e+06/ 3.2e+05
sumMemory	F	9.3e+07/ 7.7e+07	9.0e+07 /1.2e+08	6.9e+05/ 8.2e+04	2.4e+02/ 1.4e+02	1.6e+05/ 1.2e+04	2.3e+02/ 1.2e+02	2.7e+06/ 3.8e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.5e+05/ 1.1e+05	3.0e+02/ 2.2e+02	2.0e+04/ 7.6e+03	2.8e+02/ 2.0e+02	2.3e+06/ 3.1e+05
Total		9.2e+08/8.9e+08	1.1e+09/2.2e+09	3.9e+06/3.2e+06	1.6e+05/4.6e+05	8.1e+05/1.0e+05	7.8e+03/3.5e+04	2.0e+07/3.5e+06

Table 11: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.1e+05 /1.9e+07	9.1e+04 /6.9e+05	1.5e+04 /5.3e+04	2.8e+03 /4.4e+04	2.1e+06/ 1.0e+06
countJoins	F	1.4e+08 /4.3e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.3e+05	3.1e+02 /7.5e+02	1.1e+05/ 1.3e+04	2.9e+02/ 2.0e+02	2.4e+06/ 8.0e+04
filterSelectivitySkew	F	1.4e+08 /4.4e+08	2.0e+08 /7.9e+08	1.4e+06/ 8.7e+05	2.0e+02 /1.1e+03	6.5e+04/ 1.7e+04	1.8e+02 /5.0e+02	2.3e+06/ 2.3e+05
hashJoinPressure	F	1.6e+08 /4.4e+08	1.8e+08 /7.7e+08	1.6e+06/ 6.6e+05	1.6e+02 /8.2e+02	1.0e+05/ 1.6e+04	1.4e+02 /2.3e+02	2.3e+06/ 1.6e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.1e+05 /2.4e+07	7.3e+04 /3.1e+05	1.7e+04 /5.1e+04	2.8e+03 /5.0e+04	2.3e+06/ 1.1e+06
sumCost	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	2.1e+06/ 1.4e+06	4.9e+02 /1.5e+03	1.1e+05/ 2.4e+04	3.8e+02 /7.2e+02	2.2e+06/ 3.3e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.2e+06	3.6e+02 /1.2e+03	6.9e+04/ 2.1e+04	3.4e+02 /7.0e+02	2.3e+06/ 3.3e+05
sumRows	F	1.5e+08 /4.4e+08	2.3e+08 /8.3e+08	1.6e+06/ 1.2e+06	2.6e+02 /7.1e+02	5.9e+04/ 2.1e+04	2.4e+02 /5.6e+02	2.3e+06/ 3.5e+05
Total		1.3e+09/5.2e+09	1.8e+09/9.5e+09	1.1e+07/4.9e+07	1.7e+05/1.0e+06	5.4e+05/2.2e+05	7.2e+03/9.6e+04	1.8e+07/3.6e+06

Table 12: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.310	4.298	2.751	21.13	1.20$\times$	1.30$\times$	1.56$\times$	0.20 $\times$
classify Depth	F	4/7	43%	0.0445	0.0615	0.0368	0.4232	1.21$\times$	1.38$\times$	1.67$\times$	0.15 $\times$
classify tree	F	5/7	29%	0.0447	0.0615	0.0364	0.3003	1.23$\times$	1.38$\times$	1.69$\times$	0.20 $\times$
countFeatureUses	F	3/7	57%	0.0477	0.1417	0.0325	0.3786	1.46$\times$	2.97$\times$	4.35$\times$	0.37 $\times$
countLeaves	F	2/7	71%	0.0455	0.1405	0.0254	0.3690	1.79$\times$	3.09$\times$	5.54$\times$	0.38 $\times$
countNodes	F	2/7	71%	0.0458	0.1423	0.0279	0.4832	1.64$\times$	3.11$\times$	5.11$\times$	0.29 $\times$
countSmallLeaves	F	3/7	57%	0.0483	0.1357	0.0328	0.3715	1.47$\times$	2.81$\times$	4.14$\times$	0.37 $\times$
inferenceCost	F	2/7	71%	0.0484	0.1393	0.0289	0.3701	1.67$\times$	2.88$\times$	4.81$\times$	0.38 $\times$
sumImpurity	F	3/7	57%	0.0476	0.1387	0.0327	0.3813	1.45$\times$	2.92$\times$	4.24$\times$	0.36 $\times$
sumPathLengths	F	3/7	57%	0.0505	0.1382	0.0344	0.3871	1.47$\times$	2.74$\times$	4.01$\times$	0.36 $\times$
sumSamples	F	3/7	57%	0.0458	0.1349	0.0322	0.3837	1.42$\times$	2.95$\times$	4.19$\times$	0.35 $\times$
treeDepth	F	2/7	71%	0.0486	0.1389	0.0280	0.3700	1.74$\times$	2.86$\times$	4.96$\times$	0.38 $\times$
Total				3.827	5.671	3.099	25.35	1.24$\times$	1.48$\times$	1.83$\times$	0.22 $\times$
Geomean								1.47$\times$	2.41$\times$	3.54$\times$	0.30 $\times$

Table 13: Per-pass PAPI counters for **DecisionTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.4e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	1.1e+05/ 3.5e+04	1.5e+06 /2.0e+06	1.1e+05/ 3.3e+04	9.2e+08/ 1.8e+07
classify Depth	F	2.3e+08/ 1.9e+08	4.7e+08 /8.5e+08	6.6e+06/ 3.7e+05	1.6e+03/ 2.2e+02	1.9e+04 /2.8e+04	1.6e+03/ 1.8e+02	1.3e+07/ 3.3e+05
classify tree	F	2.3e+08/ 1.8e+08	4.7e+08 /8.5e+08	6.6e+06/ 3.7e+05	2.0e+03/ 2.6e+02	2.3e+04 /2.8e+04	1.9e+03/ 2.3e+02	1.3e+07/ 3.3e+05
countFeatureUses	F	2.4e+08/ 1.7e+08	7.5e+08 /7.8e+08	7.4e+06/ 1.0e+06	7.6e+02/ 4.7e+02	3.8e+04/ 1.4e+04	7.4e+02/ 4.4e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.3e+06/ 3.2e+05	1.1e+03/ 6.4e+02	4.5e+04/ 1.1e+04	1.0e+03/ 5.9e+02	1.4e+07/ 3.5e+05
countNodes	F	2.3e+08/ 1.4e+08	6.1e+08 /6.2e+08	7.1e+06/ 7.6e+05	1.5e+03/ 8.9e+02	4.7e+04/ 1.7e+04	1.3e+03/ 7.9e+02	1.4e+07/ 2.3e+05
countSmallLeaves	F	2.5e+08/ 1.7e+08	7.1e+08 /8.4e+08	5.8e+06/ 2.0e+05	7.4e+02/ 6.8e+02	3.0e+04/ 1.3e+04	7.2e+02/ 6.6e+02	1.6e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.2e+05	1.1e+03/ 6.5e+02	2.0e+04/ 8.0e+03	1.1e+03/ 6.0e+02	1.4e+07/ 4.1e+05
sumImpurity	F	2.4e+08/ 1.7e+08	7.6e+08 /7.8e+08	7.6e+06/ 1.0e+06	1.1e+03/ 7.1e+02	3.7e+04/ 1.3e+04	1.1e+03/ 6.7e+02	1.5e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.5e+06/ 1.2e+06	2.3e+03/ 6.9e+02	3.6e+04/ 1.4e+04	2.2e+03/ 6.8e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.3e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.2e+06/ 5.1e+05	7.6e+02/ 6.7e+02	3.9e+04/ 1.4e+04	7.4e+02/ 6.4e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.5e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.1e+05	1.2e+03/ 6.6e+02	1.6e+04/ 8.1e+03	1.2e+03/ 6.1e+02	1.4e+07/ 4.2e+05
Total		1.8e+10/1.6e+10	4.0e+10/6.8e+10	5.4e+08/3.3e+07	1.3e+05/4.2e+04	1.8e+06/2.1e+06	1.3e+05/3.9e+04	1.1e+09/3.0e+07

Table 14: Per-pass PAPI counters for **DecisionTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.1e+11	5.1e+10 /1.9e+11	1.8e+07 /4.5e+07	2.4e+04 /1.3e+05	3.5e+05 /2.1e+06	2.4e+04/ 2.0e+04	8.6e+08/ 1.9e+07
classify Depth	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.6e+05 /6.8e+05	4.4e+02 /3.0e+03	5.1e+03 /3.2e+04	4.2e+02 /9.0e+02	1.2e+07/ 3.2e+05
classify tree	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.6e+05 /6.1e+05	3.0e+02 /1.8e+03	5.0e+03 /3.0e+04	2.9e+02 /4.4e+02	1.2e+07/ 3.5e+05
countFeatureUses	F	7.2e+08 /1.9e+09	1.3e+09 /3.1e+09	2.1e+06 /3.2e+06	4.6e+02 /2.3e+03	5.7e+03 /6.9e+04	4.3e+02 /1.1e+03	9.4e+06/ 1.2e+06
countLeaves	F	7.1e+08 /1.9e+09	1.2e+09 /2.9e+09	4.4e+06/ 6.7e+05	1.3e+03 /2.7e+03	6.2e+03 /3.2e+04	1.3e+03/ 3.8e+02	9.2e+06/ 3.8e+05
countNodes	F	7.2e+08 /1.9e+09	1.2e+09 /3.0e+09	4.8e+06/ 6.9e+05	9.6e+02 /4.4e+03	1.7e+04 /3.3e+04	8.2e+02 /9.9e+02	9.2e+06/ 3.7e+05
countSmallLeaves	F	6.9e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	6.4e+02 /2.3e+03	4.0e+03 /7.4e+04	5.9e+02 /8.6e+02	9.3e+06/ 1.2e+06
inferenceCost	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.1e+05	1.1e+03 /1.8e+03	5.3e+03 /3.1e+04	1.1e+03/ 4.4e+02	9.2e+06/ 3.8e+05
sumImpurity	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	2.3e+06 /3.5e+06	2.1e+03/ 5.7e+02	7.4e+03 /7.5e+04	2.1e+03/ 4.7e+02	9.5e+06/ 1.2e+06
sumPathLengths	F	7.0e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	7.5e+02 /3.1e+03	4.3e+03 /7.9e+04	7.2e+02 /1.1e+03	9.2e+06/ 1.2e+06
sumSamples	F	6.8e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	1.3e+03 /2.4e+03	4.3e+03 /7.7e+04	1.2e+03/ 9.4e+02	9.3e+06/ 1.2e+06
treeDepth	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.5e+06/ 6.2e+05	6.9e+02 /2.3e+03	5.3e+03 /3.1e+04	6.4e+02 /6.8e+02	9.2e+06/ 4.0e+05
Total		2.9e+10/1.3e+11	6.4e+10/2.2e+11	4.4e+07/6.6e+07	3.4e+04/1.5e+05	4.2e+05/2.7e+06	3.4e+04/2.8e+04	9.7e+08/2.7e+07

Table 15: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0370	0.0832	0.0213	0.0912	1.74 $\times$	2.25 $\times$	3.91 $\times$	0.91 $\times$
computeWidths	M	14/15	7%	0.0297	0.0303	0.0444	0.0883	0.67 $\times$	1.02 $\times$	0.68 $\times$	0.34 $\times$
count styled	F	3/15	80%	0.0352	0.0697	0.0117	0.0867	3.02 $\times$	1.98 $\times$	5.98 $\times$	0.80 $\times$
find max Bottom	F	5/15	67%	0.0351	0.0701	0.0184	0.0859	1.90 $\times$	2.00 $\times$	3.80 $\times$	0.82 $\times$
scaleLayout	M	15/15	0%	0.0288	0.0305	0.0371	0.0881	0.78 $\times$	1.06 $\times$	0.82 $\times$	0.35 $\times$
sumTextWidth	F	3/15	80%	0.0349	0.0702	0.0116	0.0866	3.02 $\times$	2.01 $\times$	6.06 $\times$	0.81 $\times$
Total				0.2007	0.3539	0.1444	0.5269	1.39 $\times$	1.76 $\times$	2.45 $\times$	0.67 $\times$
Geomean								1.58 $\times$	1.64 $\times$	2.59 $\times$	0.62 $\times$

Table 16: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.9e+08/ 9.8e+07	2.9e+08 /4.9e+08	2.6e+06/ 1.3e+06	1.1e+03/ 4.8e+02	1.7e+05/ 2.8e+04	9.6e+02/ 3.7e+02	1.4e+07/ 1.8e+06
computeWidths	M	6.5e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	5.1e+04 /6.8e+04	4.4e+03 /1.0e+04	1.5e+03 /1.6e+04	1.3e+06/ 4.6e+05
count styled	F	1.8e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.1e+06/ 2.5e+04	8.0e+02/ 1.3e+02	3.0e+05/ 5.0e+03	7.5e+02/ 1.3e+02	1.4e+07/ 7.1e+05
find max Bottom	F	1.8e+08/ 9.4e+07	2.5e+08 /4.1e+08	2.4e+06/ 1.4e+06	4.6e+02/ 1.3e+02	1.5e+05/ 9.6e+03	4.5e+02/ 1.1e+02	1.4e+07/ 1.4e+06
scaleLayout	M	6.0e+07 /1.0e+08	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	4.6e+04 /5.0e+04	2.5e+04/ 9.0e+03	2.8e+03 /1.5e+04	1.4e+06/ 4.5e+05
sumTextWidth	F	1.8e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 1.9e+04	7.7e+02/ 1.6e+02	4.2e+05/ 4.3e+03	7.4e+02/ 1.6e+02	1.4e+07/ 7.2e+05
Total		8.4e+08/5.5e+08	1.2e+09/2.2e+09	1.2e+07/6.5e+06	1.0e+05/1.2e+05	1.1e+06/6.6e+04	7.2e+03/3.2e+04	5.7e+07/5.6e+06

Table 17: Per-pass PAPI counters for **DomTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	4.2e+08 /4.6e+08	4.4e+08 /1.8e+09	1.3e+07/ 2.8e+06	5.6e+02 /1.5e+03	4.3e+05/ 4.9e+04	4.3e+02 /1.3e+03	9.3e+06/ 1.4e+06
computeWidths	M	6.8e+07 /3.5e+08	1.5e+08 /9.0e+08	1.3e+06 /3.4e+06	4.6e+04 /2.4e+05	3.7e+03 /1.9e+04	1.3e+03 /2.2e+04	1.2e+06/ 4.7e+05
count styled	F	3.5e+08 /4.4e+08	4.3e+08 /1.7e+09	9.9e+06/ 1.2e+06	1.7e+02 /5.0e+02	3.8e+04/ 3.6e+04	1.6e+02 /4.5e+02	1.1e+07/ 4.6e+05
find max Bottom	F	3.6e+08 /4.4e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.4e+06	1.5e+02 /7.7e+02	2.7e+04/ 2.5e+04	1.3e+02 /7.6e+02	1.0e+07/ 1.0e+06
scaleLayout	M	6.8e+07 /3.6e+08	1.1e+08 /7.6e+08	9.6e+05 /4.2e+06	5.4e+04 /8.4e+04	8.7e+03 /1.0e+04	1.2e+03 /2.2e+04	1.3e+06/ 5.1e+05
sumTextWidth	F	3.6e+08 /4.4e+08	3.9e+08 /1.7e+09	1.0e+07/ 1.3e+06	2.7e+02 /1.1e+03	8.7e+04/ 3.9e+04	2.6e+02 /8.9e+02	1.0e+07/ 4.5e+05
Total		1.6e+09/2.5e+09	2.0e+09/8.6e+09	4.6e+07/1.4e+07	1.0e+05/3.3e+05	5.9e+05/1.8e+05	3.4e+03/4.8e+04	4.3e+07/4.3e+06

Table 18: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0280	0.0403	0.0273	0.1235	1.03 $\times$	1.44 $\times$	1.48 $\times$	0.33 $\times$
countInRange tight_box	F	13/17	24%	0.0273	0.0397	0.0195	0.1276	1.40 $\times$	1.45 $\times$	2.03 $\times$	0.31 $\times$
photonMappingBenchmark	F	12/17	29%	0.0418	0.0472	0.0412	0.2594	1.02 $\times$	1.13 $\times$	1.15 $\times$	0.18 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0244	0.0382	0.0164	0.1176	1.49 $\times$	1.56 $\times$	2.33 $\times$	0.32 $\times$
sumMassInRange	F	14/17	18%	0.0267	0.0398	0.0199	0.1273	1.34 $\times$	1.49 $\times$	2.00 $\times$	0.31 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0283	0.0389	0.0206	0.1172	1.38 $\times$	1.38 $\times$	1.89 $\times$	0.33 $\times$
Total				0.1764	0.2442	0.1449	0.8727	1.22 $\times$	1.38 $\times$	1.69 $\times$	0.28 $\times$
Geomean								1.26 $\times$	1.40 $\times$	1.77 $\times$	0.29 $\times$

Table 19: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.6e+06/ 1.7e+06	5.6e+02/ 5.1e+02	4.7e+03 /2.0e+04	4.4e+02/ 4.1e+02	7.8e+06/ 1.5e+06
countInRange tight_box	F	1.4e+08/ 9.9e+07	4.0e+08 /5.5e+08	2.1e+06/ 1.5e+06	1.9e+02/ 1.4e+02	3.6e+03 /1.7e+04	1.7e+02/ 1.3e+02	7.8e+06/ 1.7e+06
photonMappingBenchmark	F	2.1e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.4e+06	4.0e+02 /5.9e+02	3.6e+03 /1.7e+04	3.8e+02 /5.7e+02	5.7e+06/ 1.5e+06
pointCloudNeighborhood	F	1.2e+08/ 8.3e+07	3.3e+08 /4.8e+08	2.4e+06/ 8.1e+05	1.8e+02/ 1.4e+02	4.7e+03 /1.6e+04	1.6e+02/ 1.2e+02	7.8e+06/ 1.4e+06
sumMassInRange	F	1.4e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.5e+06/ 2.1e+06	2.0e+02 /2.1e+02	4.3e+03 /2.0e+04	1.6e+02 /1.9e+02	7.8e+06/ 1.9e+06
twoPointCorrelation bin_8_16	F	1.4e+08/ 1.0e+08	4.8e+08 /6.1e+08	2.1e+06/ 5.9e+05	2.0e+02/ 1.6e+02	3.3e+03 /1.3e+04	1.8e+02/ 1.4e+02	7.8e+06/ 1.3e+06
Total		8.9e+08/7.0e+08	3.1e+09/4.0e+09	1.4e+07/9.1e+06	1.7e+03/1.8e+03	2.4e+04/1.0e+05	1.5e+03/1.6e+03	4.5e+07/9.2e+06

Table 20: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.0e+08 /6.2e+08	4.9e+08 /1.3e+09	4.9e+06 /5.5e+06	5.1e+02 /2.5e+03	3.2e+04 /3.6e+04	3.9e+02 /2.3e+03	5.8e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.5e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	1.9e+02 /2.4e+03	5.2e+03 /2.2e+04	1.6e+02 /2.3e+03	5.8e+06/ 1.7e+06
photonMappingBenchmark	F	2.4e+08 /6.8e+08	1.1e+09 /2.0e+09	2.9e+06 /6.0e+06	5.0e+02 /4.2e+03	2.9e+03 /3.1e+04	4.9e+02 /3.0e+03	5.2e+06/ 1.5e+06
pointCloudNeighborhood	F	1.9e+08 /6.0e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.7e+06	3.0e+02 /1.3e+03	5.7e+03 /1.7e+04	2.8e+02 /1.3e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.5e+08	4.7e+08 /1.3e+09	5.1e+06 /6.3e+06	2.7e+02 /1.7e+03	4.8e+03 /2.2e+04	2.6e+02 /1.7e+03	5.7e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /5.9e+08	6.1e+08 /1.4e+09	4.8e+06/ 3.7e+06	3.2e+02 /1.6e+03	6.4e+03 /1.8e+04	2.9e+02 /1.5e+03	5.9e+06/ 1.4e+06
Total		1.2e+09/3.8e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	2.1e+03/1.4e+04	5.7e+04/1.5e+05	1.9e+03/1.2e+04	3.4e+07/9.0e+06

Table 21: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0308	0.1337	0.0064	<i>OOM</i>	4.81 $\times$	4.34 $\times$	20.91 $\times$	-
Total				0.0308	0.1337	0.0064	-	4.81 $\times$	4.34 $\times$	20.91 $\times$	-
Geomean								4.81 $\times$	4.34 $\times$	20.91 $\times$	-

Table 22: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.4e+08/ 2.6e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	6.4e+02/ 3.2e+02	2.8e+05/ 6.7e+04	5.2e+02/ 1.8e+02	1.2e+07/ 1.1e+06
Total		1.4e+08/2.6e+07	9.0e+07/1.0e+08	1.5e+06/1.2e+06	6.4e+02/3.2e+02	2.8e+05/6.7e+04	5.2e+02/1.8e+02	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for **LinearListReduction** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.7e+08/-	2.5e+08/-	4.0e+06/-	8.9e+02/-	4.0e+05/-	7.5e+02/-	2.7e+07/-
Total		6.7e+08/-	2.5e+08/-	4.0e+06/-	8.9e+02/-	4.0e+05/-	7.5e+02/-	2.7e+07/-

Table 24: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.3754	<i>OOM</i>	0.3391	<i>OOM</i>	1.11 $\times$	-	-	-
length List	F	1/2	50%	0.0419	<i>OOM</i>	0.0215	<i>OOM</i>	1.94 $\times$	-	-	-
sumList	F	2/2	0%	0.0440	<i>OOM</i>	0.0453	<i>OOM</i>	0.97 $\times$	-	-	-
sumList tail recursive	F	2/2	0%	0.0443	<i>OOM</i>	0.0445	<i>OOM</i>	1.00 $\times$	-	-	-
Total				0.5056	-	0.4504	-	1.12 $\times$	-	-	-
Geomean								1.20 $\times$	-	-	-

Table 25: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.8e+08 /9.7e+08	1.9e+09 /2.8e+09	8.5e+06/ 2.7e+06	2.8e+05 /5.7e+05	3.8e+04 /6.9e+04	1.3e+04 /1.4e+04	9.1e+06/ 7.7e+06
length List	F	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.1e+05/ 1.1e+05	2.2e+02/ 1.0e+02	2.6e+03 /1.4e+04	2.1e+02/ 8.6e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	7.1e+05/ 5.4e+05	1.8e+02/ 1.3e+02	3.0e+03 /6.8e+04	1.5e+02/ 1.2e+02	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.2e+05 /5.7e+05	2.0e+02/ 1.7e+02	2.9e+03 /5.8e+04	1.7e+02/ 1.6e+02	1.3e+07/ 1.2e+07
Total		1.2e+09/1.5e+09	4.5e+09/5.6e+09	9.9e+06/3.9e+06	2.8e+05/5.7e+05	4.7e+04/2.1e+05	1.4e+04/1.4e+04	4.8e+07/3.3e+07

Table 26: Per-pass PAPI counters for **List** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		-/-	-/-	-/-	-/-	-/-	-/-	-/-

Table 27: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0924	0.1016	0.0803	0.1209	1.15 $\times$	1.10 $\times$	1.26 $\times$	0.84 $\times$
sumTree	F	3/3	0%	0.0092	0.0410	0.0113	0.0810	0.82 $\times$	4.45 $\times$	3.63 $\times$	0.51 $\times$
sumTree TailRec	F	3/3	0%	0.0087	0.0402	0.0113	0.1963	0.77 $\times$	4.65 $\times$	3.57 $\times$	0.20 $\times$
Total				0.1103	0.1829	0.1029	0.3982	1.07 $\times$	1.66 $\times$	1.78 $\times$	0.46 $\times$
Geomean								0.90 $\times$	2.83 $\times$	2.54 $\times$	0.44 $\times$

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.4e+05 /7.2e+05	4.0e+04 /4.8e+04	2.7e+03 /1.2e+04	2.4e+03 /2.8e+03	4.5e+05/ 4.3e+05
sumTree	F	4.7e+07 /5.7e+07	2.2e+08 /2.6e+08	1.3e+04 /4.7e+04	2.5e+02/ 2.3e+02	6.3e+02 /4.1e+03	2.1e+02/2.1e+02	8.8e+05/ 6.8e+05
sumTree TailRec	F	4.4e+07 /5.7e+07	1.9e+08 /2.2e+08	6.5e+03 /2.4e+04	2.3e+02 /3.3e+02	7.5e+02 /4.1e+03	2.0e+02 /3.0e+02	8.8e+05/ 6.8e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.6e+05/7.9e+05	4.0e+04/4.9e+04	4.1e+03/2.0e+04	2.8e+03/3.3e+03	2.2e+06/1.8e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.1e+08	7.1e+08 /1.2e+09	2.3e+05 /1.4e+06	3.4e+04 /5.2e+04	3.2e+03 /2.6e+04	1.5e+03 /2.5e+03	4.3e+05/ 3.5e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	7.6e+04 /1.3e+06	2.7e+02/ 2.5e+02	1.3e+03 /2.4e+04	2.3e+02/ 2.2e+02	3.7e+05/ 3.5e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.6e+04 /1.1e+06	3.2e+02 /1.1e+03	1.1e+03 /1.9e+04	2.8e+02 /8.2e+02	3.7e+05/ 3.7e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	3.7e+05/3.9e+06	3.4e+04/5.4e+04	5.5e+03/6.9e+04	2.0e+03/3.6e+03	1.2e+06/1.1e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0125	0.0451	0.0110	0.0464	1.13 $\times$	3.62 $\times$	4.10 $\times$	0.97 $\times$
countMarked	F	3/5	40%	0.0126	0.0446	0.0150	0.0467	0.84 $\times$	3.54 $\times$	2.98 $\times$	0.95 $\times$
countSurvivors	F	4/5	20%	0.0135	0.0453	0.0130	0.0484	1.04 $\times$	3.36 $\times$	3.49 $\times$	0.93 $\times$
deadBytes	F	4/5	20%	0.0132	0.0442	0.0125	0.0474	1.05 $\times$	3.36 $\times$	3.53 $\times$	0.93 $\times$
liveBytes	F	4/5	20%	0.0130	0.0435	0.0124	0.0471	1.05 $\times$	3.35 $\times$	3.50 $\times$	0.92 $\times$
sumObjIds	F	3/5	40%	0.0125	0.0432	0.0105	0.0483	1.19 $\times$	3.45 $\times$	4.11 $\times$	0.90 $\times$
sweepUnmarked	M	5/5	0%	0.0748	0.0890	0.0802	0.1878	0.93 $\times$	1.19 $\times$	1.11 $\times$	0.47 $\times$
totalHeapSize	F	3/5	40%	0.0125	0.0652	0.0201	0.0641	0.62 $\times$	5.19 $\times$	3.24 $\times$	1.02 $\times$
touchHotObjects	M	5/5	0%	0.0734	0.0879	0.0805	0.1914	0.91 $\times$	1.20 $\times$	1.09 $\times$	0.46 $\times$
Total				0.2380	0.5080	0.2552	0.7277	0.93 $\times$	2.13 $\times$	1.99 $\times$	0.70 $\times$
Geomean								0.96 $\times$	2.85 $\times$	2.73 $\times$	0.81 $\times$

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.3e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.3e+06/ 5.9e+04	3.0e+02 /4.9e+02	3.3e+03 /3.8e+03	3.0e+02 /4.7e+02	3.0e+06/ 7.4e+05
countMarked	F	6.4e+07/ 5.6e+07	2.6e+08 /2.8e+08	1.3e+06/ 4.9e+04	3.8e+02 /4.0e+02	3.4e+03 /4.0e+03	3.5e+02 /3.7e+02	3.0e+06/ 6.8e+05
countSurvivors	F	6.8e+07/ 6.6e+07	3.1e+08 /3.8e+08	9.0e+05/ 7.2e+04	3.2e+02/ 2.4e+02	2.2e+03 /5.4e+03	3.0e+02/ 2.4e+02	2.9e+06/ 1.3e+06
deadBytes	F	6.7e+07/ 6.3e+07	2.7e+08 /3.5e+08	7.6e+05/ 5.6e+04	2.6e+02 /3.7e+02	2.3e+03 /5.5e+03	2.1e+02 /3.4e+02	2.9e+06/ 1.3e+06
liveBytes	F	6.6e+07/ 6.3e+07	2.7e+08 /3.5e+08	9.3e+05/ 5.7e+04	2.9e+02 /3.7e+02	2.4e+03 /5.8e+03	2.8e+02 /3.4e+02	2.9e+06/ 1.3e+06
sumObjIds	F	6.4e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.2e+06/ 7.5e+04	3.0e+02/ 3.0e+02	2.7e+03 /4.4e+03	2.8e+02 /2.8e+02	2.9e+06/ 7.0e+05
sweepUnmarked	M	1.9e+08 /2.3e+08	5.5e+08 /8.8e+08	1.1e+06/ 4.8e+05	9.1e+04 /1.1e+05	6.9e+03 /1.1e+04	5.7e+03 /6.8e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.3e+07/ 5.4e+07	2.5e+08 /2.7e+08	1.3e+06/ 6.0e+05	6.1e+02 /6.6e+02	2.5e+03 /1.3e+04	4.7e+02 /5.6e+02	3.0e+06/ 5.6e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	1.1e+06/ 4.8e+05	1.0e+05 /1.3e+05	5.2e+03 /1.1e+04	4.5e+03 /6.6e+03	1.9e+06/ 1.0e+06
Total		8.3e+08/8.8e+08	3.0e+09/4.0e+09	9.9e+06/1.9e+06	2.0e+05/2.4e+05	3.1e+04/6.4e+04	1.2e+04/1.6e+04	2.5e+07/8.6e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.5e+05/ 4.6e+05	5.0e+02/ 1.9e+02	5.0e+03 /7.9e+03	4.8e+02/ 1.7e+02	1.8e+06/ 4.3e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.7e+05	2.2e+02 /2.4e+02	5.0e+03 /9.4e+03	1.8e+02 /2.0e+02	1.8e+06/ 4.4e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.8e+05 /3.9e+05	2.2e+02 /5.0e+02	1.9e+03 /6.6e+03	1.9e+02 /4.3e+02	1.8e+06/ 7.4e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.8e+05 /4.1e+05	2.4e+02 /4.4e+02	3.6e+03 /6.4e+03	2.1e+02 /4.0e+02	1.8e+06/ 7.6e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.5e+05 /4.0e+05	1.6e+02 /4.2e+02	1.9e+03 /6.5e+03	1.2e+02 /3.9e+02	1.8e+06/ 7.6e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.4e+05/ 5.1e+05	3.5e+02 /3.6e+02	4.6e+03 /9.6e+03	3.2e+02 /3.3e+02	1.8e+06/ 4.2e+05
sweepUnmarked	M	2.7e+08 /7.7e+08	7.6e+08 /2.1e+09	1.8e+05 /2.4e+06	8.0e+04 /1.4e+05	3.5e+03 /1.9e+04	3.0e+03 /6.2e+03	1.8e+06/ 9.3e+05
totalHeapSize	F	2.4e+08 /2.4e+08	4.4e+08 /7.8e+08	2.5e+06/ 5.3e+05	5.5e+02 /9.3e+02	4.2e+04/ 8.9e+03	3.6e+02 /6.6e+02	1.4e+06/ 4.2e+05
touchHotObjects	M	2.7e+08 /7.9e+08	7.7e+08 /2.2e+09	2.3e+05 /1.8e+06	8.6e+04 /1.2e+05	3.1e+03 /2.1e+04	3.2e+03 /4.9e+03	1.7e+06/ 9.4e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	5.3e+06/7.3e+06	1.7e+05/2.6e+05	7.0e+04/9.5e+04	8.1e+03/1.4e+04	1.6e+07/5.9e+06

Table 33: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1199	0.1301	0.1311	0.4073	0.91 $\times$	1.08 $\times$	0.99 $\times$	0.32 $\times$
autorefineMaxLevel	F	4/8	50%	0.0207	0.0453	0.0193	0.2346	1.07 $\times$	2.19 $\times$	2.34 $\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0197	0.0460	0.0115	0.2344	1.71 $\times$	2.33 $\times$	3.99 $\times$	0.20 $\times$
diffPW	M	8/8	0%	0.1207	0.1321	0.1291	0.4090	0.93 $\times$	1.09 $\times$	1.02 $\times$	0.32 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0215	0.0470	0.0184	0.2364	1.17 $\times$	2.19 $\times$	2.55 $\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0257	0.0626	0.0231	0.1996	1.11 $\times$	2.44 $\times$	2.71 $\times$	0.31 $\times$
pmapCutHistogram	F	4/8	50%	0.0319	0.0556	0.0127	0.2420	2.52 $\times$	1.75 $\times$	4.39 $\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0201	0.1752	0.0116	0.2023	1.74 $\times$	8.71 $\times$	15.16 $\times$	0.87 $\times$
Total				0.3801	0.6938	0.3568	2.166	1.07 $\times$	1.83 $\times$	1.94 $\times$	0.32 $\times$
Geomean								1.31 $\times$	2.17 $\times$	2.85 $\times$	0.29 $\times$

Table 34: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.3e+08	5.7e+08 /1.2e+09	1.5e+06 /1.8e+06	2.6e+05 /2.6e+05	1.3e+04 /2.5e+04	7.4e+03 /2.3e+04	4.6e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08/ 9.8e+07	2.5e+08 /3.7e+08	2.5e+06/ 2.1e+05	8.0e+02/ 1.5e+02	1.4e+04/ 6.5e+03	7.9e+02/ 1.2e+02	6.1e+06/ 1.0e+06
compressMass	F	1.0e+08/ 5.8e+07	2.5e+08 /2.8e+08	2.1e+06/ 2.6e+04	6.0e+02/ 3.5e+02	1.9e+04/ 5.2e+03	6.1e+02/ 3.4e+02	6.3e+06/ 6.8e+05
diffPW	M	2.8e+08 /3.2e+08	5.9e+08 /1.3e+09	6.2e+05 /1.7e+06	1.4e+05 /2.0e+05	9.8e+03 /2.2e+04	7.6e+03 /2.0e+04	4.2e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.3e+07	3.3e+08 /4.9e+08	2.4e+06/ 3.1e+05	5.0e+02/ 1.8e+02	1.8e+04/ 9.2e+03	4.9e+02/ 1.7e+02	6.3e+06/ 1.6e+06
norm2Estimate	F	1.3e+08/ 8.8e+07	3.4e+08 /4.4e+08	1.0e+06/ 4.5e+05	9.1e+02/ 4.0e+02	1.4e+04/ 1.0e+04	8.3e+02/ 3.1e+02	6.2e+06/ 1.3e+06
pmapCutHistogram	F	1.6e+08/ 6.4e+07	2.8e+08 /3.7e+08	1.7e+06/ 6.1e+04	1.1e+03/ 3.7e+02	1.0e+05/ 7.5e+03	1.1e+03/ 3.5e+02	4.8e+06/ 1.3e+06
truncateTolViolations	F	1.0e+08/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 2.9e+04	8.3e+02/ 3.0e+02	1.8e+04/ 4.9e+03	8.0e+02/ 2.8e+02	6.4e+06/ 7.0e+05
Total		1.3e+09/1.1e+09	2.8e+09/4.7e+09	1.4e+07/4.7e+06	4.0e+05/4.6e+05	2.1e+05/9.0e+04	2.0e+04/4.4e+04	4.5e+07/1.0e+07

Table 35: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	3.2e+05 /5.0e+06	1.3e+05 /3.3e+05	7.4e+03 /4.3e+04	5.3e+03 /3.1e+04	4.2e+06/ 2.0e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	2.9e+05 /2.1e+06	4.0e+02 /1.2e+03	1.7e+03 /4.0e+04	3.8e+02 /4.2e+02	4.2e+06/ 7.1e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.3e+06	3.3e+02 /1.6e+03	1.8e+03 /3.0e+04	2.9e+02 /2.9e+02	4.1e+06/ 4.0e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.3e+09	3.1e+05 /4.9e+06	1.2e+05 /2.5e+05	6.2e+03 /4.2e+04	5.5e+03 /2.4e+04	4.0e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.6e+05 /2.6e+06	3.8e+02 /4.7e+02	1.8e+03 /4.1e+04	3.7e+02 /4.0e+02	4.2e+06/ 1.0e+06
norm2Estimate	F	2.9e+08 /1.0e+09	5.1e+08 /1.3e+09	1.5e+06 /2.5e+06	8.2e+02 /1.5e+03	2.9e+04 /4.1e+04	6.2e+02 /1.1e+03	3.8e+06/ 9.9e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.4e+05 /2.0e+06	9.2e+02 /2.0e+03	1.0e+04 /3.7e+04	8.7e+02/ 5.3e+02	3.9e+06/ 7.2e+05
truncateTolViolations	F	2.2e+08 /1.0e+09	4.4e+08 /1.3e+09	3.4e+05 /1.3e+06	2.0e+03/ 1.8e+03	4.3e+03 /2.7e+04	1.9e+03/ 6.8e+02	2.9e+06/ 4.1e+05
Total		2.2e+09/1.0e+10	4.5e+09/1.4e+10	4.0e+06/2.2e+07	2.6e+05/5.9e+05	6.2e+04/3.0e+05	1.5e+04/5.9e+04	3.1e+07/8.0e+06

Table 36: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0847	0.1125	0.0898	0.3450	0.94 $\times$	1.33$\times$	1.25$\times$	0.33 $\times$
sum tree	F	5/5	0%	0.0277	0.0499	0.0270	0.2731	1.02 $\times$	1.80$\times$	1.85$\times$	0.18 $\times$
Total				0.1124	0.1624	0.1169	0.6181	0.96 $\times$	1.44$\times$	1.39$\times$	0.26 $\times$
Geomean								0.98 $\times$	1.55$\times$	1.52$\times$	0.24 $\times$

Table 37: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.5e+05 /1.1e+06	6.6e+04 /1.2e+05	3.8e+03 /1.3e+04	2.3e+03 /3.0e+03	1.6e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.4e+08	4.6e+08 /5.4e+08	8.8e+04 /2.9e+05	1.1e+02 /1.8e+02	7.9e+02 /9.2e+03	8.8e+01 /1.6e+02	1.8e+06/ 1.3e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	5.3e+05/1.4e+06	6.6e+04/1.2e+05	4.6e+03/2.2e+04	2.4e+03/3.1e+03	3.4e+06/2.4e+06

Table 38: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.3e+08 /1.6e+09	9.5e+08 /2.4e+09	4.2e+05 /2.8e+06	6.8e+04 /1.7e+05	6.5e+03 /5.3e+04	3.4e+03 /4.6e+03	1.4e+06/ 8.9e+05
sum tree	F	2.5e+08 /1.4e+09	5.8e+08 /1.0e+09	6.0e+04 /2.3e+06	1.4e+02 /1.3e+03	1.4e+03 /4.2e+04	1.2e+02 /5.8e+02	1.4e+06/ 9.6e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	4.8e+05/5.1e+06	6.8e+04/1.7e+05	8.0e+03/9.5e+04	3.5e+03/5.2e+03	2.8e+06/1.9e+06

Table 39: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0146	0.0275	0.0130	0.1079	1.12$\times$	1.88$\times$	2.12$\times$	0.25 $\times$
countLazyNodes	F	4/10	60%	0.0125	0.0278	0.0082	0.1109	1.53$\times$	2.22$\times$	3.40$\times$	0.25 $\times$
countTerminals	F	3/10	70%	0.0120	0.0327	0.0156	0.0989	0.77 $\times$	2.73$\times$	2.10$\times$	0.33 $\times$
decayTrieStats	M	10/10	0%	0.0759	0.0823	0.0908	0.2433	0.84 $\times$	1.09 $\times$	0.91 $\times$	0.34 $\times$
resetTraversalState	M	10/10	0%	0.0764	0.0819	0.0853	0.2638	0.89 $\times$	1.07 $\times$	0.96 $\times$	0.31 $\times$
sumPrefixFreq	F	3/10	70%	0.0121	0.0466	0.0193	0.1118	0.63 $\times$	3.85$\times$	2.41$\times$	0.42 $\times$
sumSubtreeHints	F	3/10	70%	0.0117	0.0265	0.0121	0.1054	0.97 $\times$	2.26$\times$	2.18$\times$	0.25 $\times$
Total				0.2152	0.3253	0.2443	1.042	0.88 $\times$	1.51$\times$	1.33$\times$	0.31 $\times$
Geomean								0.93 $\times$	1.97$\times$	1.83$\times$	0.30 $\times$

Table 40: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.4e+07/ 3.9e+07	1.3e+08 /2.1e+08	1.5e+06/ 4.9e+05	1.6e+02/ 1.1e+02	5.4e+04/ 4.8e+03	1.5e+02/ 1.0e+02	3.7e+06/ 5.6e+05
countLazyNodes	F	6.3e+07/ 3.2e+07	1.3e+08 /1.6e+08	1.3e+06/ 2.6e+04	1.7e+02 /1.9e+02	1.5e+04/ 3.3e+03	1.5e+02 /1.6e+02	4.1e+06/ 6.0e+05
countTerminals	F	6.0e+07/ 2.8e+07	1.0e+08 /1.2e+08	1.2e+06/ 1.0e+04	3.1e+02/ 2.4e+02	2.7e+04/ 2.3e+03	3.0e+02/ 2.1e+02	4.0e+06/ 2.4e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	7.2e+05 /1.4e+06	9.6e+04 /1.5e+05	5.7e+03 /2.2e+04	3.3e+03 /2.6e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.1e+08	2.8e+08 /7.0e+08	1.0e+06/ 8.9e+05	9.1e+04 /1.4e+05	9.2e+03 /1.5e+04	3.5e+03 /1.6e+04	3.3e+06/ 9.1e+05
sumPrefixFreq	F	6.1e+07/ 2.8e+07	1.1e+08 /1.2e+08	1.2e+06/ 1.4e+04	6.4e+02/ 5.0e+02	2.7e+04/ 2.6e+03	4.6e+02/ 3.4e+02	4.0e+06/ 2.4e+05
sumSubtreeHints	F	5.9e+07/ 2.8e+07	1.1e+08 /1.2e+08	1.4e+06/ 1.3e+04	2.3e+02/ 1.7e+02	2.6e+04/ 2.5e+03	2.1e+02/ 1.4e+02	4.0e+06/ 2.8e+05
Total		6.5e+08/5.8e+08	1.2e+09/2.2e+09	8.3e+06/2.8e+06	1.9e+05/2.8e+05	1.6e+05/5.2e+04	8.0e+03/4.4e+04	2.6e+07/4.0e+06

Table 41: Per-pass PAPI counters for **Trie** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.5e+08	2.3e+08 /7.5e+08	1.1e+06 /1.5e+06	2.3e+02 /1.0e+03	3.0e+03 /2.7e+04	1.9e+02 /8.4e+02	2.8e+06/ 4.9e+05
countLazyNodes	F	1.4e+08 /5.6e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	1.0e+02 /8.8e+02	2.9e+03 /2.2e+04	8.4e+01 /4.0e+02	2.7e+06/ 3.5e+05
countTerminals	F	1.7e+08 /5.0e+08	2.0e+08 /7.3e+08	4.1e+06/ 6.5e+05	1.9e+02 /1.1e+03	1.0e+05/ 1.6e+04	1.8e+02 /2.8e+02	2.5e+06/ 2.0e+05
decayTrieStats	M	1.9e+08 /1.0e+09	4.5e+08 /2.1e+09	3.0e+05 /5.1e+06	8.2e+04 /3.9e+05	4.3e+03 /3.2e+04	3.5e+03 /3.8e+04	2.9e+06/ 1.3e+06
resetTraversalState	M	2.0e+08 /1.1e+09	3.7e+08 /2.0e+09	2.7e+05 /3.4e+06	1.1e+05 /1.7e+05	5.0e+03 /3.2e+04	4.2e+03 /3.4e+04	3.0e+06/ 1.0e+06
sumPrefixFreq	F	1.5e+08 /5.4e+08	2.1e+08 /7.3e+08	4.2e+06/ 7.0e+05	5.4e+02 /1.3e+03	8.7e+04/ 1.7e+04	4.2e+02/ 3.5e+02	2.4e+06/ 1.9e+05
sumSubtreeHints	F	1.3e+08 /5.4e+08	2.1e+08 /7.3e+08	1.5e+06/ 6.1e+05	1.7e+02 /9.4e+02	4.2e+03 /1.6e+04	1.6e+02 /2.6e+02	2.8e+06/ 2.0e+05
Total		1.1e+09/4.8e+09	1.9e+09/7.8e+09	1.4e+07/1.3e+07	2.0e+05/5.6e+05	2.1e+05/1.6e+05	8.7e+03/7.4e+04	1.9e+07/3.7e+06